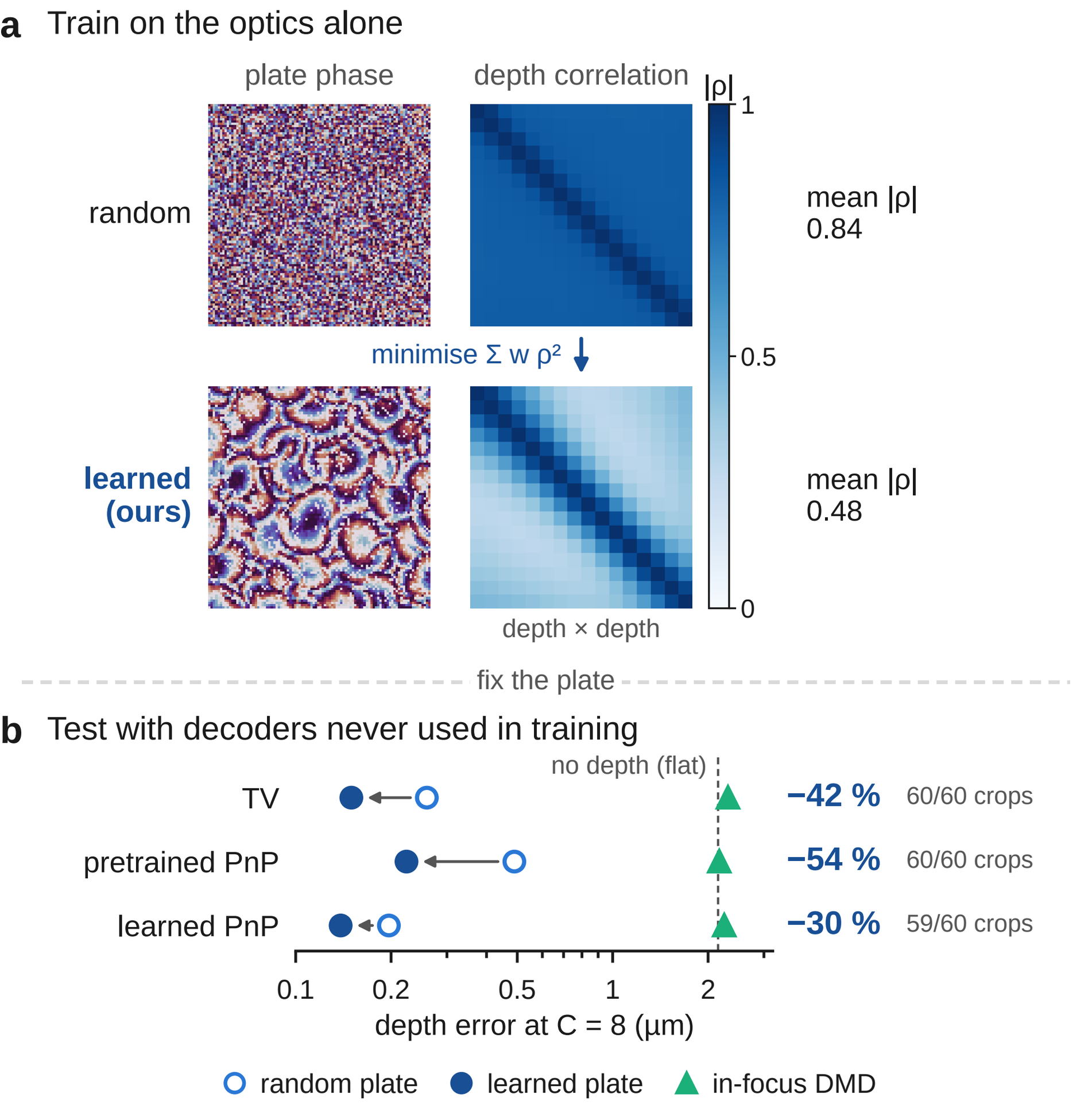

a
Train on the optics alone
plate phase
depth correlation
|ρ|
1
0.5
0
random
mean |ρ|
0.84
minimise Σ w ρ²
learned
(ours)
mean |ρ|
0.48
depth × depth
fix the plate
b
Test with decoders never used in training
no depth (flat)
TV
pretrained PnP
learned PnP
−42 %
60/60 crops
−54 %
60/60 crops
−30 %
59/60 crops
0.1
0.2
0.5
1
2
depth error at C = 8 (μm)
random plate
learned plate
in-focus DMD